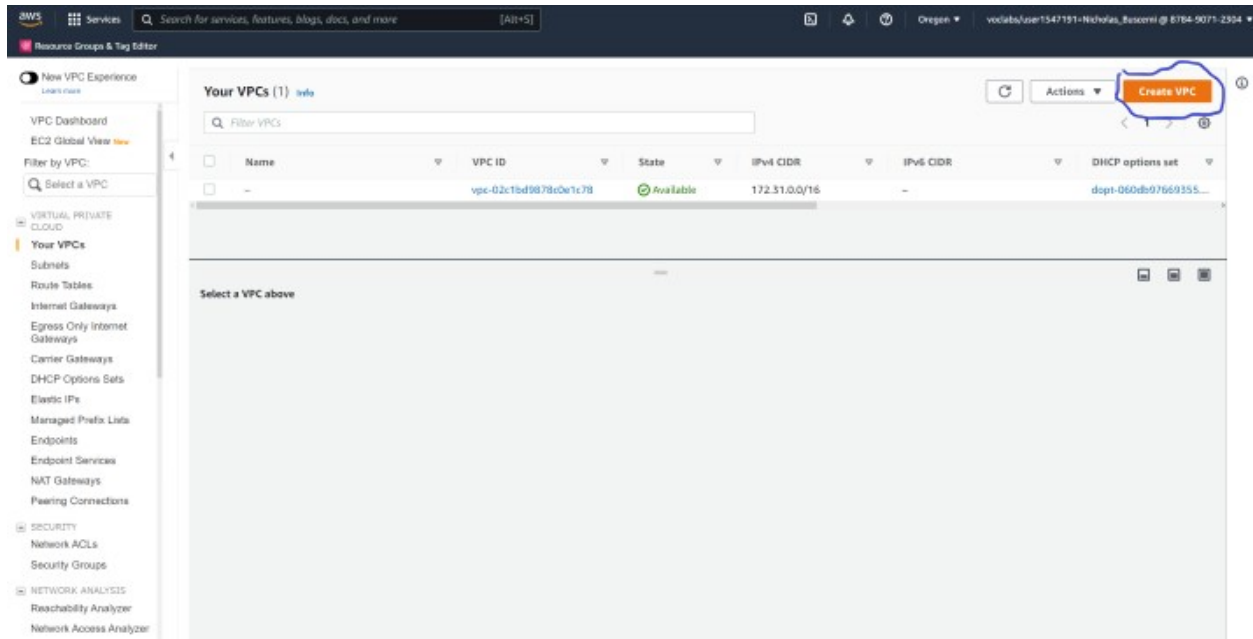


## Step 1: Create a VPC and Subnets as well as routing and security groups

- Go to “Your VPCs” from the VPC service on the AWS management console and click on the orange “Create VPC” button



- Only create a vpc here and give it a name. You are free to make your own name or follow along with the one put here

- Give it a 192.168.0.0/16 CIDR block and leave everything else as default. Click create.

## Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

### VPC settings

Resources to create [Info](#)  
Create only the VPC resource or create VPC, subnets, etc.

☒ VPC only ☐ VPC, subnets, etc.

Name tag - *optional*  
Creates a tag with a key of 'Name' and a value that you specify.

Demo VPC

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

192.168.0.0/16

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Default

### Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

| Key      | Value - <i>optional</i> |        |
|----------|-------------------------|--------|
| Q Name X | Q Demo VPC X            | Remove |

Add new tag

You can add 49 more tags.

Cancel **Create VPC**

- To create your subnets go to Subnets on the left hand side of the VPC service and click on it
- Add your VPC ID to where it asks

## Create subnet [Info](#)

### VPC

#### VPC ID

Create subnets in this VPC.

vpc-03bd2b389d2a4d45e (Demo VPC) ▼

#### Associated VPC CIDRs

##### IPv4 CIDRs

192.168.0.0/16

- Assign it a name letting you know what it is your first public subnet
- Put it in any availability zone and give it a CIDR of 192.168.1.0/24

### Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

#### Subnet 1 of 1

##### Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public Subnet

The name can be up to 256 characters long.

##### Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US West (Oregon) / us-west-2a ▼

##### IPv4 CIDR block [Info](#)

Q 192.168.1.0/24 X

##### ▼ Tags - optional

##### Key

Q Name X

##### Value - optional

Q Public Subnet X

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

- Add a second subnet and name it Private Subnet 1 or something to let you know it is your first private subnet
- Put it in the same availability zone as the first subnet you made and give it a CIDR of 192.168.2.0/24

Subnet 2 of 2

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private Subnet 1

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US West (Oregon) / us-west-2a

IPv4 CIDR block [Info](#)

192.168.2.0/24

Tags - optional

Key

Value - optional

Name

Private Subnet 1

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

- Add a third subnet and assign a name letting you know it is the second private subnet you will be making
- Put it in the same availability zone as your first public subnet and give it a CIDR of 192.168.3.0/24

### Subnet 3 of 3

#### Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private Subnet 2

The name can be up to 256 characters long.

#### Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US West (Oregon) / us-west-2a

#### IPv4 CIDR block [Info](#)

192.168.3.0/24

#### ▼ Tags - *optional*

Key

Name

Value - *optional*

Private Subnet 2

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

- Add a fourth and final subnet and give it a name letting you know it is the third private subnet
- Put it in a different availability zone from the rest of your subnets and give it a CIDR of 192.168.4.0/24

#### Subnet 4 of 4

##### Subnet name

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

##### Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

##### IPv4 CIDR block [Info](#)

##### ▼ Tags - *optional*

Key

Value - *optional*

Remove

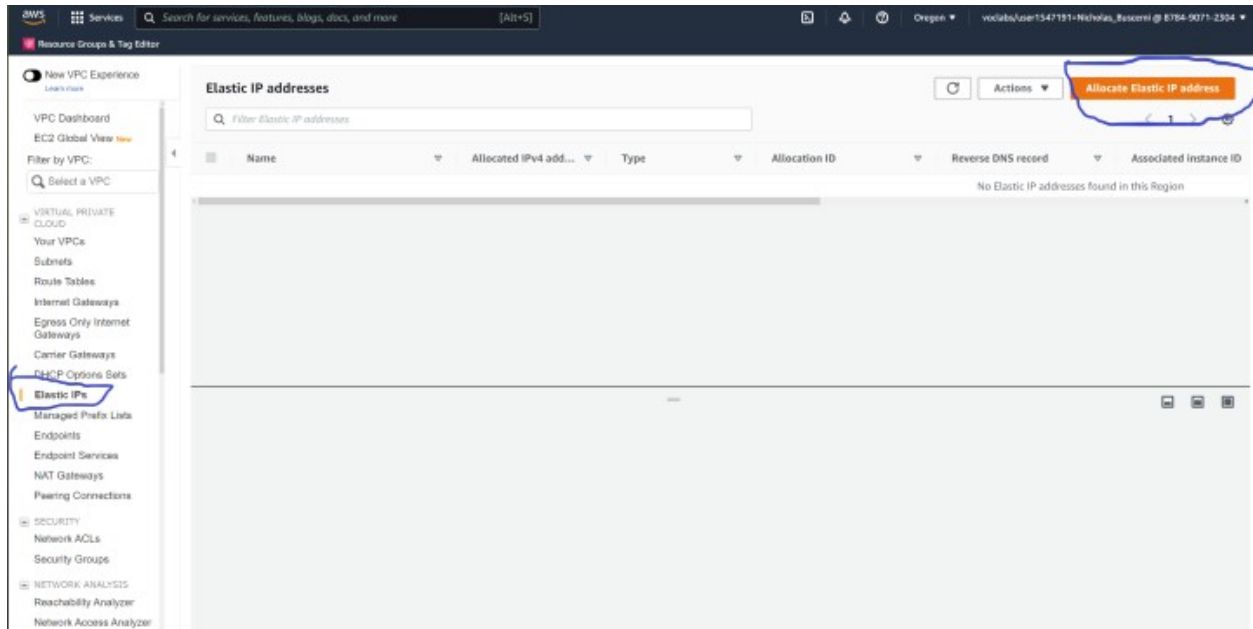
Add new tag

You can add 49 more tags.

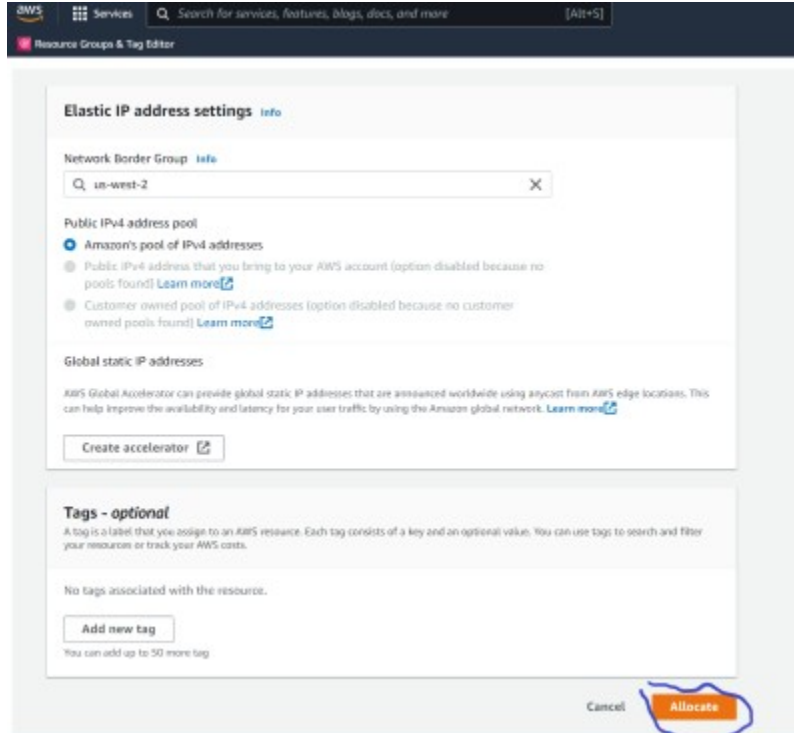
Remove

Add new subnet

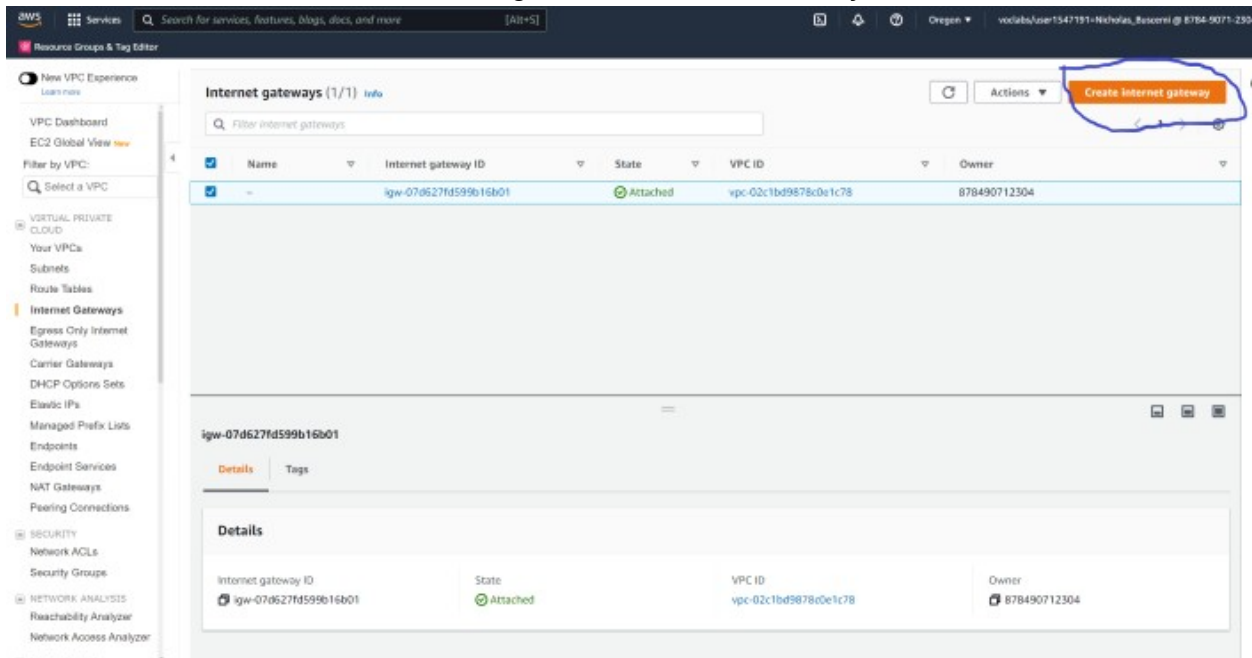
- Set up for route tables
- Allocate an Elastic IP address by going to Elastic IPs on the left hand side and click “Allocate Elastic IP address”



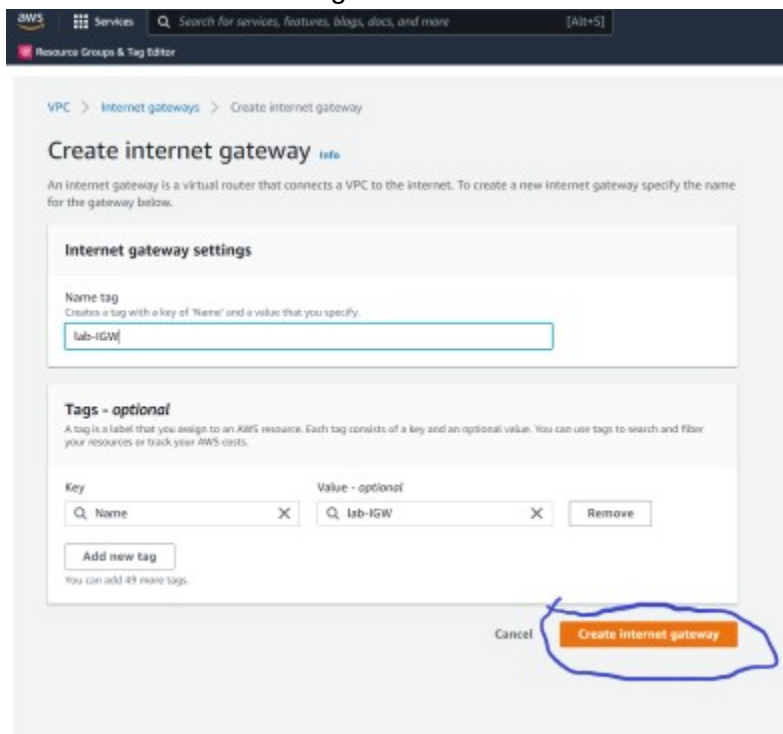
- Everything should be good as default but make sure that you are in the same region you have been creating everything in and then press “Allocate”. You can also add a name tag if you wish but it isn't necessary



- Now create an internet gateway and attach it to the VPC by going to Internet Gateways on the left hand side and clicking “Create Internet Gateway”

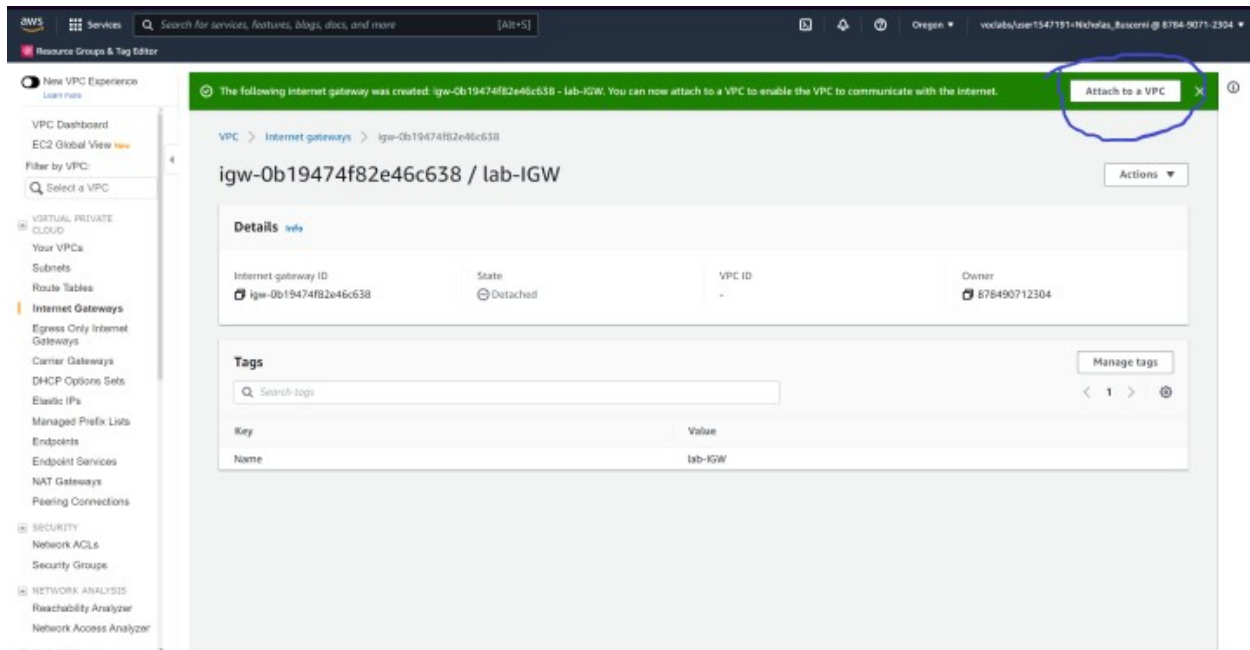


- Name it something similar to what is below and then click “Create Internet Gateway”

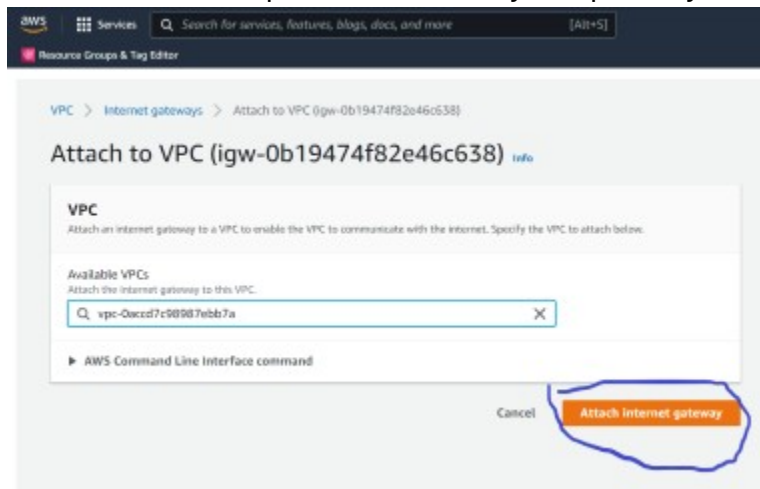




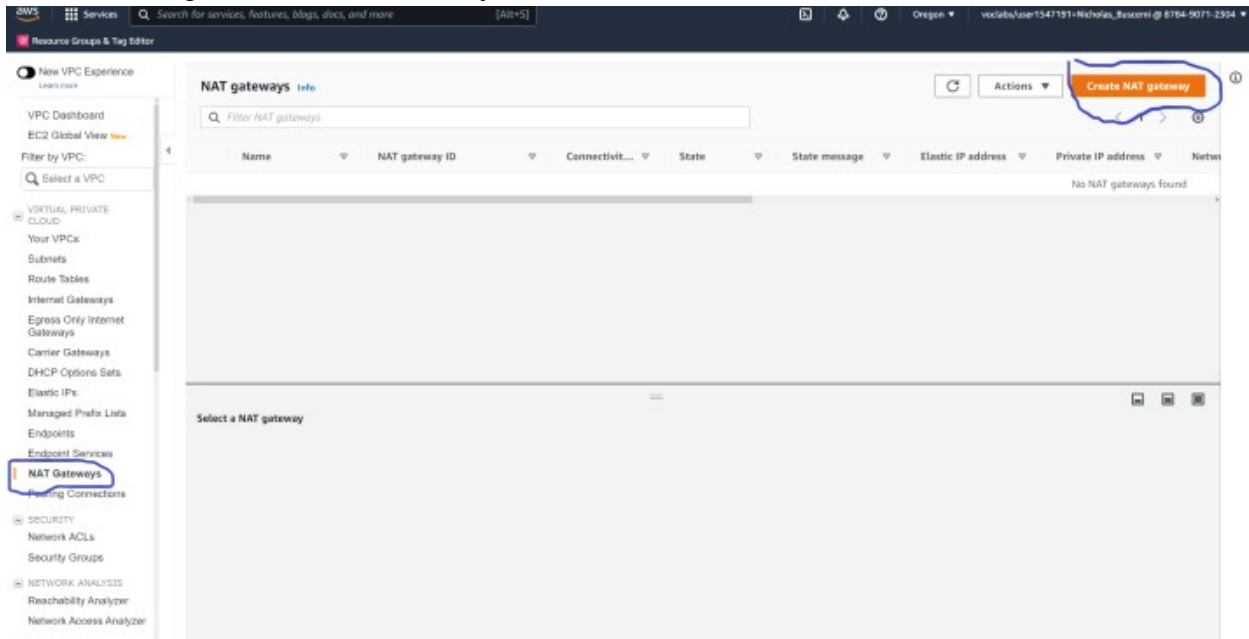
- Once it is created attach it to your VPC by clicking “Attach to a VPC” on the top of the screen



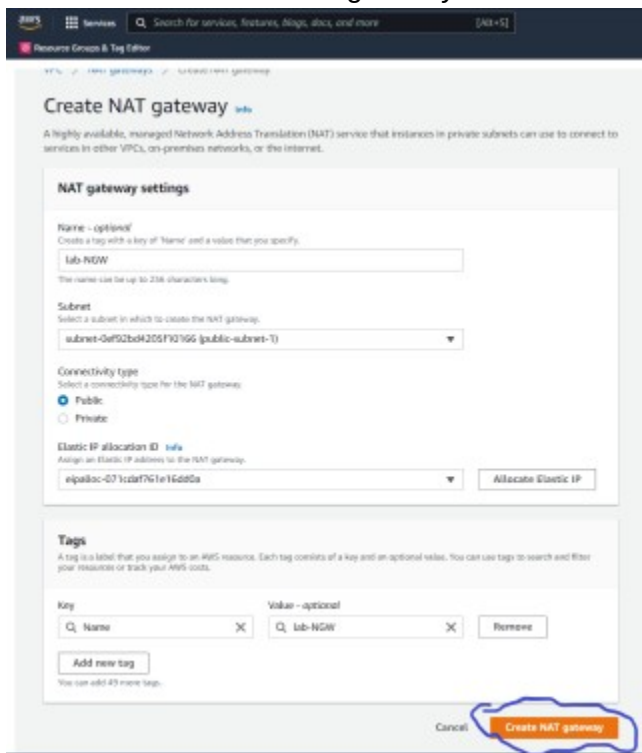
- Click the drop down and select your vpc that you made



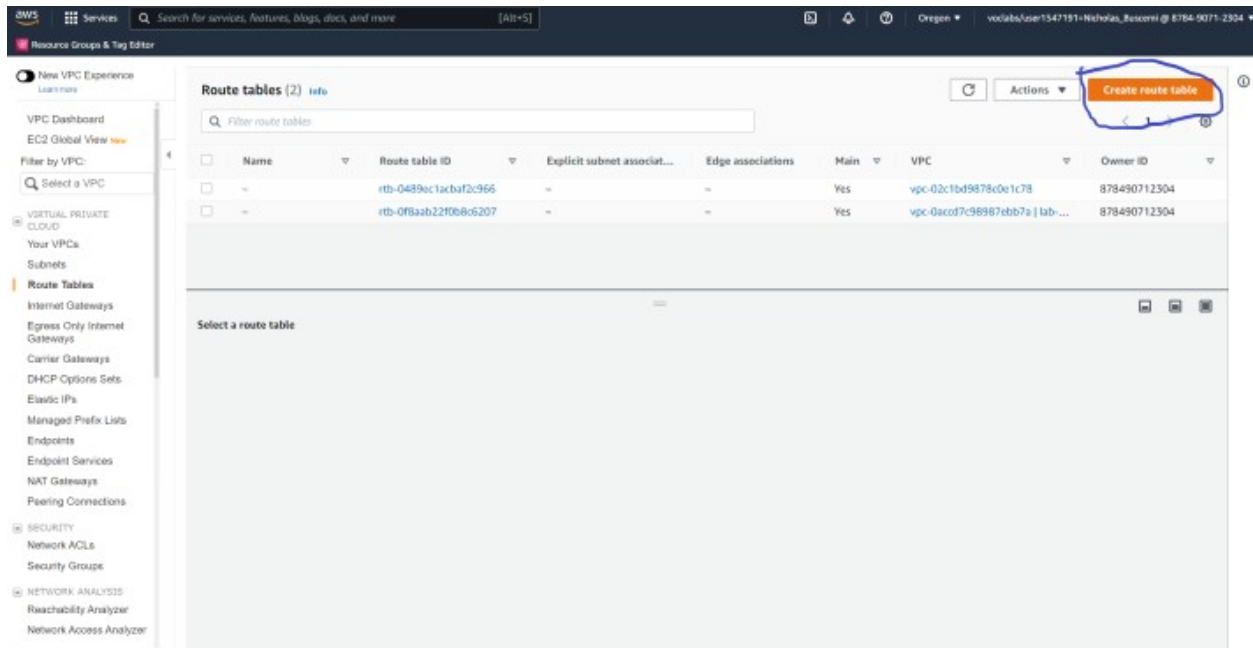
- Create a NAT Gateway by clicking on Nat Gateways on the left hand side and then clicking “Create NAT Gateway”



- Give it a name similar to the one below and assign it to a public subnet
- Click the drop down for Elastic IPs and click the one you created previously
- Click “Create NAT gateway”



- Create Route Tables by first heading to “Route Tables” on the left hand side
- Click “Create route table”



- Give it a name letting you know this is the public route table for your lab
- Assign your VPC to it and click “Create route table”

### Create route table info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

#### Route table settings

**Name - optional**  
Create a tag with a key of 'Name' and a value that you specify.

lab-public-table

**VPC**  
The VPC to use for this route table.

vpc-0accd7c98987ebb7a (lab-VPC)

#### Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

**Key** **Value - optional**

Q Name X Q lab-public-table X Remove

Add new tag

You can add 49 more tags.

Cancel **Create route table**

- Make a second route table naming it something to let you know that this is the private route table for your lab and assign your VPC to it

**Create route table** info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

**Route table settings**

**Name - optional**  
Create a tag with a key of 'Name' and a value that you specify.  
lab-private-table

**VPC**  
The VPC to use for this route table.  
vpc-0accd7c98987ebb7a (lab-VPC)

**Tags**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

**Key** Name **Value - optional** lab-private-table **Remove**

**Add new tag**  
You can add 49 more tags.

**Create route table**

- Now associate your subnets with their respective route table
- Click on the public route table and click on "Subnet association" next to "Details"

**Route tables (1/4)** info

Filter route tables

| Name                                                 | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                            | Owner ID     |
|------------------------------------------------------|-----------------------|---------------------------|-------------------|------|--------------------------------|--------------|
| -                                                    | rtb-0489ec1ach72c966  | -                         | -                 | Yes  | vpc-02c1bd9878e0e1c78          | 878490712304 |
| <input checked="" type="checkbox"/> lab-public-table | rtb-096cb259306549323 | -                         | -                 | No   | vpc-0accd7c98987ebb7a   lab... | 878490712304 |
| <input type="checkbox"/> lab-private-table           | rtb-0f53c7f17d08253bb | -                         | -                 | No   | vpc-0accd7c98987ebb7a   lab... | 878490712304 |
| <input type="checkbox"/> -                           | rtb-0f8a2b22f0b805207 | -                         | -                 | Yes  | vpc-0accd7c98987ebb7a   lab... | 878490712304 |

**rtb-096cb259306549323 / lab-public-table**

**Details** **Routes** **Subnet associations** **Edge associations** **Route propagation** **Tags**

**Details**

Route table ID: rtb-096cb259306549323  
Main: No  
Owner ID: 878490712304  
VPC: vpc-0accd7c98987ebb7a | lab-VPC

Explicit subnet associations: -  
Edge associations: -

**Run Reachability Analyzer**

- Click on “Edit subnet associations”

The screenshot shows the AWS Management Console interface for the 'Route tables (1/4)' page. The left sidebar contains navigation options like 'VPC Dashboard', 'EC2 Global View', and 'Subnets'. The main content area displays a table of route tables. The 'lab-public-table' is selected, and its details are shown below. The 'Explicit subnet associations (0)' section is highlighted with a blue circle, and the 'Edit subnet associations' button is circled in blue.

| Name              | Route table ID        | Explicit subnet associat... | Edge associations | Main | VPC                             | Owner ID     |
|-------------------|-----------------------|-----------------------------|-------------------|------|---------------------------------|--------------|
| lab-public-table  | rtb-095cb259306549323 | -                           | -                 | No   | vpc-0accd7c98987ebb7a   lab-... | 878490712304 |
| lab-private-table | rtb-0f53cf717d08253bb | -                           | -                 | No   | vpc-0accd7c98987ebb7a   lab-... | 878490712304 |
| -                 | rtb-0f8aab22f0b806207 | -                           | -                 | Yes  | vpc-0accd7c98987ebb7a   lab-... | 878490712304 |

- Click on your public subnet and then click “Save associations”

## Edit subnet associations

Change which subnets are associated with this route table.

The screenshot shows the 'Edit subnet associations' dialog box. It contains a table of available subnets and a section for selected subnets.

| Name                 | Subnet ID                       | IPv4 CIDR             | IPv6 CIDR | Route table ID                    |
|----------------------|---------------------------------|-----------------------|-----------|-----------------------------------|
| Private Subnet 3     | subnet-092c5783f28705ad7        | 192.168.4.0/24        | -         | Main (rtb-010a67db84e50ef4        |
| <b>Public Subnet</b> | <b>subnet-06bc254b8826ccb04</b> | <b>192.168.1.0/24</b> | <b>-</b>  | <b>Main (rtb-010a67db84e50ef4</b> |
| Private Subnet 1     | subnet-0b2f703b31b1dbf78        | 192.168.2.0/24        | -         | Main (rtb-010a67db84e50ef4        |
| Private Subnet 2     | subnet-00fbb6995bc7ecbc6        | 192.168.3.0/24        | -         | Main (rtb-010a67db84e50ef4        |

**Selected subnets**

subnet-06bc254b8826ccb04 / Public Subnet X

Cancel Save associations

- Now add a route to our public route table to get access to the internet gateway

- Click on “Routes” next to “Details” and click “Edit routes”

New VPC Experience [Learn more](#)

VPC Dashboard  
EC2 Global View **New**

Filter by VPC:

- VIRTUAL PRIVATE CLOUD
  - Your VPCs
  - Subnets
  - Route Tables**
  - Internet Gateways
  - Egress Only Internet Gateways
  - Carrier Gateways
  - DHCP Option Sets
  - Elastic IPs
  - Managed Prefix Lists
  - Endpoints
  - Endpoint Services
  - NAT Gateways
  - Peering Connections
- SECURITY
  - Network ACLs
  - Security Groups
- NETWORK ANALYSIS
  - Reachability Analyzer
  - Network Access Analyzer
- DNS FIREWALL
  - Rule Groups
  - Domain Lists
- NETWORK FIREWALL
  - Firewalls
  - Firewall Policies
  - Network Firewall Rule Groups
- VIRTUAL PRIVATE NETWORK (VPN)

You have successfully updated subnet associations for rtb-07fadf542dba89a17 / lab-public-table.

**Route tables (1/4)** [Info](#)

| <input type="checkbox"/>            | Name              | Route table ID        | Explicit subnet associat... | Edge associations | Main |
|-------------------------------------|-------------------|-----------------------|-----------------------------|-------------------|------|
| <input type="checkbox"/>            | -                 | rtb-010a67db84e50ef44 | -                           | -                 | Yes  |
| <input checked="" type="checkbox"/> | lab-public-table  | rtb-07fadf542dba89a17 | subnet-06bc254b8826c...     | -                 | No   |
| <input type="checkbox"/>            | lab-private-table | rtb-079223b43d56c7bf8 | -                           | -                 | No   |
| <input type="checkbox"/>            | -                 | rtb-0dffa298cf2ccb9c3 | -                           | -                 | Yes  |

**rtb-07fadf542dba89a17 / lab-public-table**

Details **Routes** Subnet associations Edge associations Route propagation Tags

**Routes (1)**

| Destination    | Target | Status | Propagated |
|----------------|--------|--------|------------|
| 192.168.0.0/16 | local  | Active | No         |

- Add a new route having a destination of anywhere and a target of your internet gateway and click “Save changes”

## Edit routes

### Edit routes

| Destination    | Target                             | Status   |
|----------------|------------------------------------|----------|
| 192.168.0.0/16 | <input type="text" value="local"/> | ✔ Active |
| Propagated     |                                    |          |
| No             |                                    |          |

### Edit routes

| Destination                            | Target                                             | Status |
|----------------------------------------|----------------------------------------------------|--------|
| <input type="text" value="0.0.0.0/0"/> | <input type="text" value="igw-0a2e406efd9302f16"/> | -      |
| Propagated                             |                                                    |        |
| No                                     |                                                    |        |

- Do the same thing for your private route table by clicking on it and going to its subnet associations and editing them

**Route tables (1/4)**

| Name              | Route table ID        | Explicit subnet associat... | Edge associations | Main | VPC                            | Owner ID     |
|-------------------|-----------------------|-----------------------------|-------------------|------|--------------------------------|--------------|
| -                 | rtb-0489c1acba7c966   | -                           | -                 | Yes  | vpc-02c1bd9873c0e1c78          | 878490712304 |
| lab-public-table  | rtb-096cb259306549323 | 2 subnets                   | -                 | No   | vpc-0accd7c98987ebb7a   lab... | 878490712304 |
| lab-private-table | rtb-0f53c#f17d08253bb | -                           | -                 | No   | vpc-0accd7c98987ebb7a   lab... | 878490712304 |
| -                 | rtb-0f8a8b22f0b8c6207 | -                           | -                 | Yes  | vpc-0accd7c98987ebb7a   lab... | 878490712304 |

**rtb-0f53c#f17d08253bb / lab-private-table**

Details | Routes | **Subnet associations** | Edge associations | Route propagation | Tags

**Explicit subnet associations (0)**

Find subnet association

| Subnet ID                                                          | IPv4 CIDR | IPv6 CIDR |
|--------------------------------------------------------------------|-----------|-----------|
| No subnet associations<br>You do not have any subnet associations. |           |           |

**Subnets without explicit associations (2)**

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table.

- Click on all three of your private subnets and save the associations

## Edit subnet associations

Change which subnets are associated with this route table.

**Available subnets (3/4)**

Filter subnet associations

| Name             | Subnet ID                | IPv4 CIDR      | IPv6 CIDR | Route table ID                         |
|------------------|--------------------------|----------------|-----------|----------------------------------------|
| Private Subnet 3 | subnet-092c5783f28705ad7 | 192.168.4.0/24 | -         | Main (rtb-010a67db84e50ef44)           |
| Public Subnet    | subnet-06bc254b8826ccb04 | 192.168.1.0/24 | -         | rtb-07adf542dba89a17 / lab-pu<br>table |
| Private Subnet 1 | subnet-0b2f703b31b1dbf78 | 192.168.2.0/24 | -         | Main (rtb-010a67db84e50ef44)           |
| Private Subnet 2 | subnet-00fbb6995bc7ecbc6 | 192.168.3.0/24 | -         | Main (rtb-010a67db84e50ef44)           |

**Selected subnets**

subnet-00fbb6995bc7ecbc6 / Private Subnet 2 X subnet-0b2f703b31b1dbf78 / Private Subnet 1 X subnet-092c5783f28705ad7 / Private Subnet 3 X

Cancel Save associations

- Go to edit the routes of the private table



New VPC Experience

Learn more

VPC Dashboard

EC2 Global View New

Filter by VPC:

Select a VPC

VIRTUAL PRIVATE CLOUD

Your VPCs

Subnets

**Route Tables**

Internet Gateways

Egress Only Internet Gateways

Carrier Gateways

DHCP Option Sets

Elastic IPs

Managed Prefix Lists

Endpoints

Endpoint Services

NAT Gateways

Peering Connections

SECURITY

Network ACLs

Security Groups

NETWORK ANALYSIS

Reachability Analyzer

Network Access Analyzer

DNS FIREWALL

Rule Groups

Domain Lists

NETWORK FIREWALL

Firewalls

Firewall Policies

Network Firewall Rule Groups

VIRTUAL PRIVATE NETWORK (VPN)

You have successfully updated subnet associations for rtb-079223b43d56c7bf8 / lab-private-table.

Route tables (1/4) Info

Actions

Create route table

Filter route tables

< 1 >

|                                     | Name              | Route table ID        | Explicit subnet associat... | Edge associations | Main |
|-------------------------------------|-------------------|-----------------------|-----------------------------|-------------------|------|
| <input type="checkbox"/>            | -                 | rtb-010a67db84e50ef44 | -                           | -                 | Yes  |
| <input type="checkbox"/>            | lab-public-table  | rtb-07fadf542dba89a17 | subnet-06bc254b8826c...     | -                 | No   |
| <input checked="" type="checkbox"/> | lab-private-table | rtb-079223b43d56c7bf8 | 3 subnets                   | -                 | No   |
| <input type="checkbox"/>            | -                 | rtb-0dffa298cf2ccb9c3 | -                           | -                 | Yes  |

rtb-079223b43d56c7bf8 / lab-private-table

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (1)

Edit routes

Filter routes

Both

< 1 >

| Destination    | Target | Status | Propagated |
|----------------|--------|--------|------------|
| 192.168.0.0/16 | local  | Active | No         |

- Add a route to the private table that has a destination of anywhere and a target of your Nat gateway that you made earlier

## Edit routes

**Edit routes**

| Destination    | Target                             | Status   |
|----------------|------------------------------------|----------|
| 192.168.0.0/16 | <input type="text" value="local"/> | ✓ Active |
| Propagated     |                                    |          |
| No             |                                    |          |

**Edit routes**

| Destination                            | Target                                             | Status |
|----------------------------------------|----------------------------------------------------|--------|
| <input type="text" value="0.0.0.0/0"/> | <input type="text" value="nat-0dc88ba1b12f5d4bf"/> | -      |
| Propagated                             |                                                    |        |
| No                                     |                                                    |        |
| <input type="button" value="Remove"/>  |                                                    |        |

- Now to create our security groups (One for our bastion host, web server, app server, and our database) we will head to Security Groups on the left and click “Create security group”

The screenshot shows the AWS Management Console interface for the 'Security Groups' page. The left sidebar contains navigation links for VPC Dashboard, EC2 Global View, and various VPC resources. The main content area displays a table of existing security groups. A blue circle highlights the 'Create security group' button in the top right corner of the main content area.

| Name | Security group ID    | Security group name | VPC ID                | Description                | Owner        | Inbound rules       |
|------|----------------------|---------------------|-----------------------|----------------------------|--------------|---------------------|
| -    | sg-0050135137115f782 | default             | vpc-0ac0d7c58987ebb7a | default VPC security gr... | 878490712304 | 1 Permission ent... |
| -    | sg-0a4022b171aa06442 | default             | vpc-02c1bd9878c0e1c78 | default VPC security gr... | 878490712304 | 1 Permission ent... |

- Give it a name and description letting you know it is for a bastion host
- Assign your VPC to it

- Give it three inbound rules, one for SSH using your IP and one for HTTP using 0.0.0.0/0 as well as https using 0.0.0.0/0

VPC > Security Groups > Create security group

## Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

### Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

### Inbound rules [Info](#)

| Type <a href="#">Info</a> | Protocol <a href="#">Info</a> | Port range <a href="#">Info</a> | Source <a href="#">Info</a>                                                                             | Description - optional <a href="#">Info</a> |                                       |
|---------------------------|-------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------|
| HTTP ▼                    | TCP                           | 80                              | Anywh... <input type="text"/> <input type="button" value="0.0.0.0/0"/> <input type="button" value="X"/> | <input type="text"/>                        | <input type="button" value="Delete"/> |
| HTTPS ▼                   | TCP                           | 443                             | Anywh... <input type="text"/> <input type="button" value="0.0.0.0/0"/> <input type="button" value="X"/> | <input type="text"/>                        | <input type="button" value="Delete"/> |
| SSH ▼                     | TCP                           | 22                              | Anywh... <input type="text"/> <input type="button" value="0.0.0.0/0"/> <input type="button" value="X"/> | <input type="text"/>                        | <input type="button" value="Delete"/> |

- Create another security group
- Give it a name and description letting you know it is for a Web server

- Assign your VPC to it
- Give it the same inbound rules as the Bastion Host security group

VPC > Security Groups > Create security group

## Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

### Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

### Inbound rules [Info](#)

| Type <a href="#">Info</a> | Protocol <a href="#">Info</a> | Port range <a href="#">Info</a> | Source <a href="#">Info</a>                         | Description - optional <a href="#">Info</a> |                   |
|---------------------------|-------------------------------|---------------------------------|-----------------------------------------------------|---------------------------------------------|-------------------|
| HTTP ▼                    | TCP                           | 80                              | Anywh... ▼ <input type="text" value="0.0.0.0/0"/> X | <input type="text"/>                        | <div>Delete</div> |
| HTTPS ▼                   | TCP                           | 443                             | Anywh... ▼ <input type="text" value="0.0.0.0/0"/> X | <input type="text"/>                        | <div>Delete</div> |
| SSH ▼                     | TCP                           | 22                              | Anywh... ▼ <input type="text" value="0.0.0.0/0"/> X | <input type="text"/>                        | <div>Delete</div> |

Add rule

- Create another security group
- Give it a name and description letting you know it is for an app server

- Assign your VPC to it
- Give it an inbound rule for All ICMP -IPv4 with a source of your web server SG and another inbound rule for SSH with a source of your bastion host SG

VPC > Security Groups > Create security group

## Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

### Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

### Inbound rules [Info](#)

| Type <a href="#">Info</a> | Protocol <a href="#">Info</a> | Port range <a href="#">Info</a> | Source <a href="#">Info</a>             | Description - optional <a href="#">Info</a> |                        |
|---------------------------|-------------------------------|---------------------------------|-----------------------------------------|---------------------------------------------|------------------------|
| All ICMP - IPv4 ▼         | ICMP                          | All                             | Custom ▼ <input type="text" value="Q"/> | <input type="text"/>                        | <div>Del<br/>ete</div> |
|                           |                               |                                 | <div>sg-094827ea3d4c27f60</div>         |                                             |                        |
| SSH ▼                     | TCP                           | 22                              | Custom ▼ <input type="text" value="Q"/> | <input type="text"/>                        | <div>Del<br/>ete</div> |
|                           |                               |                                 | <div>sg-041812008930fcc7b</div>         |                                             |                        |
| <div>Add rule</div>       |                               |                                 |                                         |                                             |                        |

- Create one final security group
- Give it a name and description letting you know it is for a database server

- Assign your VPC to it
- Give it two inbound rules both for MYSQL/Aurora and give one of them a source of your app server SG and the other one a source of your bastion host SG

VPC > Security Groups > Create security group

## Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

### Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

### Inbound rules [Info](#)

| Type <a href="#">Info</a> | Protocol <a href="#">Info</a> | Port range <a href="#">Info</a> | Source <a href="#">Info</a>             | Description - optional <a href="#">Info</a> |                   |
|---------------------------|-------------------------------|---------------------------------|-----------------------------------------|---------------------------------------------|-------------------|
| MYSQL/Aurora ▼            | TCP                           | 3306                            | Custom ▼ <input type="text" value="Q"/> | <input type="text"/>                        | <div>Delete</div> |
|                           |                               |                                 | sg-0782f81911c052438 ✕                  |                                             |                   |
| MYSQL/Aurora ▼            | TCP                           | 3306                            | Custom ▼ <input type="text" value="Q"/> | <input type="text"/>                        | <div>Delete</div> |
|                           |                               |                                 | sg-094827ea3d4c27f60 ✕                  |                                             |                   |

Add rule

- Go back to your bastion host inbound rules and add one more for MYSQL/Aurora and a source of your database SG
- Go back to your web server inbound rules and add one more for All ICMP - IPv4 and a source of your app server SG
- Go back to your app server inbound rules and add one more for MYSQL/Aurora and a source of your database SG and then an HTTP and HTTPS rule both with a source of 0.0.0.0/0

## Step 2: Create Servers

- Create Bastion Host

- Select Amazon Linux 2 AMI

### Step 1: Choose an Amazon Machine Image (AMI)

[Cancel and Exit](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Q Search for an AMI by entering a search term e.g. "Windows" X

Search by Systems Manager parameter

Quick Start

My AMIs

AWS Marketplace

Community AMIs

☐ Free tier only ⓘ

  
Amazon Linux  
Free tier eligible

**Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type** - ami-0ca285d4c2cda3300 (64-bit x86) / ami-0f48d15c9efb5f63d (64-bit Arm)  
  
Amazon Linux 2 comes with five years support. It provides Linux kernel 5.10 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is now under maintenance only mode and has been removed from this wizard.  
  
Root device type: ebs    Virtualization type: hvm    ENA Enabled: Yes

Select

☒ 64-bit (x86)  
☐ 64-bit (Arm)

  
Amazon Linux  
Free tier eligible

**Amazon Linux 2 AMI (HVM) - Kernel 4.14, SSD Volume Type** - ami-00af37d1144686454 (64-bit x86) / ami-0d3127dab514c6a1a (64-bit Arm)  
  
Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is now under maintenance only mode and has been removed from this wizard.  
  
Root device type: ebs    Virtualization type: hvm    ENA Enabled: Yes

Select

☒ 64-bit (x86)  
☐ 64-bit (Arm)

  
macOS Monterey 12.3.1 - ami-0b24eb4af3f138c47  
  
The macOS Monterey AMI is an EBS-backed, AWS-supported image. This AMI includes the AWS Command Line Interface, Command Line Tools for Xcode, Amazon SSM Agent, and Homebrew. The AWS Homebrew Tap includes the latest versions of multiple AWS packages included in the AMI.  
  
Root device type: ebs    Virtualization type: hvm    ENA Enabled: Yes

Select

64-bit (Mac)

  
macOS Big Sur 11.6.5 - ami-04bf4e24478eb9533  
  
The macOS Big Sur AMI is an EBS-backed, AWS-supported image. This AMI includes the AWS Command Line Interface, Command Line Tools for Xcode, Amazon SSM Agent, and Homebrew. The AWS Homebrew Tap includes the latest versions of multiple AWS packages included in the AMI.  
  
Root device type: ebs    Virtualization type: hvm    ENA Enabled: Yes

Select

64-bit (Mac)

  
macOS Catalina 10.15.7 - ami-0aec57ca49edbbaaf  
  
The macOS Catalina AMI is an EBS-backed, AWS-supported image. This AMI includes the AWS Command Line Interface, Command Line Tools for Xcode, Amazon SSM Agent, and Homebrew. The AWS Homebrew Tap includes the latest versions of multiple AWS packages included in the AMI.  
  
Root device type: ebs    Virtualization type: hvm    ENA Enabled: Yes

Select

64-bit (Mac)

- Select t2.micro



## Step 2: Choose an Instance Type

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instances are virtual servers that can run applications. They have varying combinations of CPU, memory, storage, and networking capacity, and give you the flexibility to choose the appropriate mix of resources for your applications. [Learn more](#) about instance types and how they can meet your computing needs.

Filter by: All instance families Current generation [Show/Hide Columns](#)

Currently selected: t2.micro (- ECUs, 1 vCPUs, 2.5 GHz, -, 1 GiB memory, EBS only)

|                                     | Family | Type                           | vCPUs ⓘ | Memory (GiB) | Instance Storage (GB) ⓘ | EBS-Optimized Available ⓘ | Network Performance ⓘ | IPv6 Support ⓘ |
|-------------------------------------|--------|--------------------------------|---------|--------------|-------------------------|---------------------------|-----------------------|----------------|
| <input type="checkbox"/>            | t2     | t2.nano                        | 1       | 0.5          | EBS only                | -                         | Low to Moderate       | Yes            |
| <input checked="" type="checkbox"/> | t2     | t2.micro<br>Free tier eligible | 1       | 1            | EBS only                | -                         | Low to Moderate       | Yes            |
| <input type="checkbox"/>            | t2     | t2.small                       | 1       | 2            | EBS only                | -                         | Low to Moderate       | Yes            |
| <input type="checkbox"/>            | t2     | t2.medium                      | 2       | 4            | EBS only                | -                         | Low to Moderate       | Yes            |
| <input type="checkbox"/>            | t2     | t2.large                       | 2       | 8            | EBS only                | -                         | Low to Moderate       | Yes            |
| <input type="checkbox"/>            | t2     | t2.xlarge                      | 4       | 16           | EBS only                | -                         | Moderate              | Yes            |
| <input type="checkbox"/>            | t2     | t2.2xlarge                     | 8       | 32           | EBS only                | -                         | Moderate              | Yes            |
| <input type="checkbox"/>            | t3     | t3.nano                        | 2       | 0.5          | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3     | t3.micro                       | 2       | 1            | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3     | t3.small                       | 2       | 2            | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3     | t3.medium                      | 2       | 4            | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3     | t3.large                       | 2       | 8            | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3     | t3.xlarge                      | 4       | 16           | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3     | t3.2xlarge                     | 8       | 32           | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3a    | t3a.nano                       | 2       | 0.5          | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3a    | t3a.micro                      | 2       | 1            | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3a    | t3a.small                      | 2       | 2            | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3a    | t3a.medium                     | 2       | 4            | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3a    | t3a.large                      | 2       | 8            | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |
| <input type="checkbox"/>            | t3a    | t3a.xlarge                     | 4       | 16           | EBS only                | Yes                       | Up to 5 Gigabit       | Yes            |

- Put in your VPC and Public Subnet and enable auto assign public IP

## Step 3: Configure Instance Details

Configure the instance to suit your requirements. You can launch multiple instances from the same AMI, request Spot instances to take advantage of the lower pricing, a management role to the instance, and more.

**Number of instances** ⓘ  [Launch into Auto Scaling Group](#) ⓘ

---

**Purchasing option** ⓘ ☐ Request Spot instances

---

**Network** ⓘ  ⓘ [Create new VPC](#)

**Subnet** ⓘ  ⓘ [Create new subnet](#)  
250 IP Addresses available

---

**Auto-assign Public IP** ⓘ  ⓘ

---

**Hostname type** ⓘ  ⓘ

**DNS Hostname** ⓘ ☒ Enable IP name IPv4 (A record) DNS requests  
☒ Enable resource-based IPv4 (A record) DNS requests  
☐ Enable resource-based IPv6 (AAAA record) DNS requests

- Storage leave default
- Add a name tag to let you know this is the Bastion Host

## Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver.

A copy of a tag can be applied to volumes, instances or both.

Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

| Key  | Value        | Instances                           | Volumes                             | Network Interfaces                  |
|------|--------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Name | Bastion Host | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |

[Add another tag](#) (Up to 50 tags maximum)

- Select an existing group and select your Bastion Host SG

## Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

**Assign a security group:** ☐ Create a new security group  
☒ Select an existing security group

| Security Group ID                                        | Name               | Description                       | Actions                     |
|----------------------------------------------------------|--------------------|-----------------------------------|-----------------------------|
| <input type="checkbox"/> sg-0ed4d62f77158ba28            | default            | default VPC security group        | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-0782f81911c052438            | MyAppServerSG      | My App Server Security Group      | <a href="#">Copy to new</a> |
| <input checked="" type="checkbox"/> sg-041812008930fcc7b | MyBastionHostSG    | My Bastion Host Security Group    | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-0810d8978701fb97b            | MyDatabaseServerSG | My Database Server Security Group | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-094827ea3d4c27f60            | MyWebServerSG      | My Web Server Security Group      | <a href="#">Copy to new</a> |

- Launch and choose an existing keypair. This can be downloaded from the lab page
- To make the Web Server follow the same steps until you get to Step 3
- Follow along like previously and change your network, subnet, and enable auto assign public ip
- Then go to user data and type this into it to set up the web server
  - `#!/bin/bash`
  - `sudo yum update -y`

- `sudo amazon-linux-extras install -y lamp-mariadb10.2-php7.2 php7.2`
- `Sudo yum install -y httpd`
- `sudo systemctl start httpd`
- `sudo systemctl enable httpd`

User data ⓘ ☒ As text ☐ As file ☐ Input is already base64 encoded

```
#!/bin/bash
sudo yum update -y
sudo amazon-linux-extras install -y lamp-mariadb10.2-php7.2 php7.2
Sudo yum install -y httpd
sudo systemctl start httpd
sudo systemctl enable httpd
```

- Storage leave default
- Give it a name tag letting you know it is the Web Server

#### Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver.

A copy of a tag can be applied to volumes, instances or both.

Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

| Key (128 characters maximum) | Value (256 characters maximum) | Instances ⓘ                         | Volumes ⓘ                           | Network Interfaces ⓘ                |
|------------------------------|--------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Name                         | Web Server                     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |

- Select an existing security group and select your web server SG

#### Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group: ☐ Create a new security group  
☒ Select an existing security group

| Security Group ID                                        | Name               | Description                       | Actions                     |
|----------------------------------------------------------|--------------------|-----------------------------------|-----------------------------|
| <input type="checkbox"/> sg-0ed4d62f77158ba28            | default            | default VPC security group        | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-0782f81911c052438            | MyAppServerSG      | My App Server Security Group      | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-041812008930fcc7b            | MyBastionHostSG    | My Bastion Host Security Group    | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-0810d8978701fb97b            | MyDatabaseServerSG | My Database Server Security Group | <a href="#">Copy to new</a> |
| <input checked="" type="checkbox"/> sg-094827ea3d4c27f60 | MyWebServerSG      | My Web Server Security Group      | <a href="#">Copy to new</a> |

- Just like before launch and use the existing keypair

- Follow the same steps once again to create the app server until you get to step 3
- Put in your VPC and then choose Private Subnet 1 for the subnet and leave auto assign public ip disabled

### Step 3: Configure Instance Details

Configure the instance to suit your requirements. You can launch multiple instances from the same AMI, request Spot instances to take advantage of the lower pricing, assign an access management role to the instance, and more.

**Number of instances** ⓘ  [Launch into Auto Scaling Group](#) ⓘ

---

**Purchasing option** ⓘ ☐ Request Spot instances

---

**Network** ⓘ  ⓘ [Create new VPC](#)

**Subnet** ⓘ  ⓘ [Create new subnet](#)  
251 IP Addresses available

**Auto-assign Public IP** ⓘ  ⓘ

- Then go into metadata and type this out to set up a database server on our app server
  - `#!/bin/bash`
  - `sudo yum install -y mariadb-server`
  - `Sudo service mariadb start` (note that the image is incorrect, make sure to add sudo)

**User data** ⓘ ☒ As text ☐ As file ☐ Input is already base64 encoded

```
#!/bin/bash
sudo yum install -y mariadb-server
Service mariadb start
```

- Give it a name letting you know it is the App Server

| Key  | Value      | Instances                           | Volumes                             | Network Interfaces                  |
|------|------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Name | App Server | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |

- Select an existing security group and select the app server SG

### Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

**Assign a security group:** ☐ Create a new security group  
☒ Select an existing security group

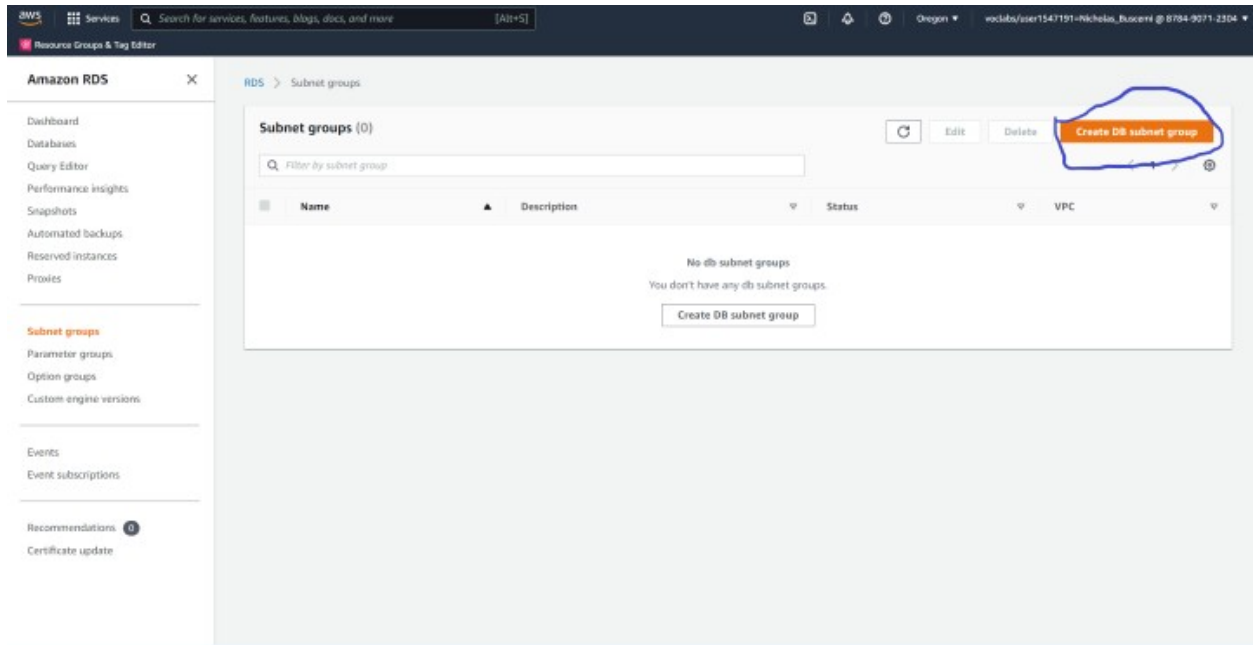
| Security Group ID                                        | Name               | Description                       | Actions                     |
|----------------------------------------------------------|--------------------|-----------------------------------|-----------------------------|
| <input type="checkbox"/> sg-0ed4d6277158ba28             | default            | default VPC security group        | <a href="#">Copy to new</a> |
| <input checked="" type="checkbox"/> sg-0782f81911c052438 | MyAppServerSG      | My App Server Security Group      | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-041812008930fcc7b            | MyBastionHostSG    | My Bastion Host Security Group    | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-0810d8978701fb97b            | MyDatabaseServerSG | My Database Server Security Group | <a href="#">Copy to new</a> |
| <input type="checkbox"/> sg-094827ea3d4c27f60            | MyWebServerSG      | My Web Server Security Group      | <a href="#">Copy to new</a> |

- Just like before launch and use the existing keypair

### Step 3: Create a Database

- Create a DB subnet group by first heading to the Amazon RDS service page on the AWS management console

- Click on Subnet Groups on the left hand side and then click on “Create DB subnet group”



- Give it a name and description letting you know what it is and then assign your VPC to it
- Put in the availability zones you used for your subnets
- Select subnets 3 and 4
- Click create

Amazon RDS

×

Dashboard

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Certificate update

RDS > Subnet groups > Create DB subnet group

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

DBSubnetGroup

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

Subnet Group for database

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

Demo VPC (vpc-03bd2b389d2a4d45e)

Add subnets

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-west-2a

us-west-2b

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

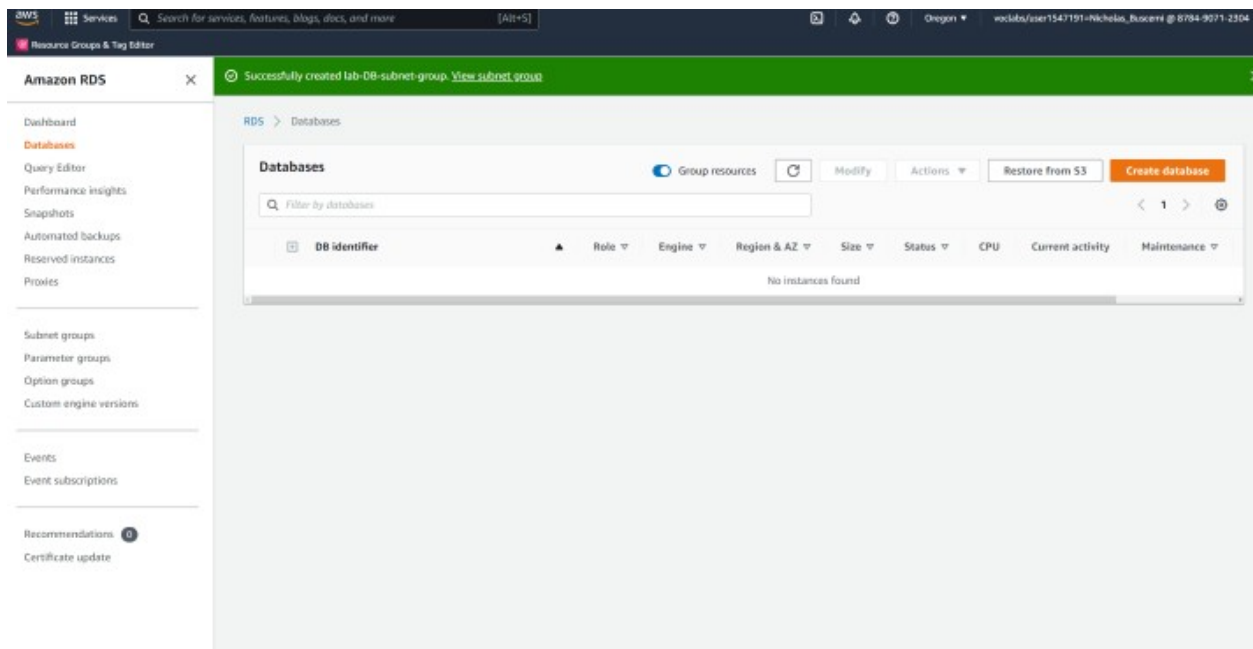
subnet-092c5783f28705ad7 (192.168.4.0/24)

subnet-00fbb6995bc7ecbc6 (192.168.3.0/24)

Subnets selected (2)

| Availability zone | Subnet ID                | CIDR block     |
|-------------------|--------------------------|----------------|
| us-west-2a        | subnet-00fbb6995bc7ecbc6 | 192.168.3.0/24 |
| us-west-2b        | subnet-092c5783f28705ad7 | 192.168.4.0/24 |

- Go to Databases on the left hand side and click on “Create Database”



- Click on Standard create and MariaDB for the engine type

# Create database

## Choose a database creation method [Info](#)

- ☒ **Standard create**  
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

- ☐ **Easy create**  
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

## Engine options

### Engine type [Info](#)

- ☐ Amazon Aurora



- ☐ MySQL



- ☒ MariaDB



- ☐ PostgreSQL



- ☐ Oracle



- ☐ Microsoft SQL Server



### Version

MariaDB 10.6.7



- Make sure you click on Free tier here



AWS Services Search for services, features, blogs, docs, and more [Alt+S]

Resource Groups & Tag Editor

### Templates

Choose a sample template to meet your use case.

☐ Production

Use defaults for high availability and fast, consistent performance.

☐ Dev/Test

This instance is intended for development use outside of a production environment.

☒ Free tier

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

### Availability and durability

Deployment options [Info](#)

The deployment options below are limited to those supported by the engine you selected above.

- ☐ Multi-AZ DB Cluster - new
- ☐ Multi-AZ DB instance (not supported for Multi-AZ DB cluster snapshot)
- ☒ Single DB instance (not supported for Multi-AZ DB cluster snapshot)

### Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

database-1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

- Give it an identifier you can easily identify it with

- Give it a master username or leave it as default admin. For the purpose of these instructions I will be using root
- Give it a password that you write down somewhere else to make sure you have the correct one. For the purpose of these instructions I will be using Re:Start!9

## Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

dbinstance

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

root

1 to 16 alphanumeric characters. First character must be a letter.

☐ Auto generate a password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

.....

Constraints: At least 8 printable ASCII characters. Can't contain any of the following: / (slash), '(single quote), "(double quote) and @ (at sign).

Confirm password [Info](#)

.....

- Everything between this and the last step is left default
- Assign your vpc

- Make sure your subnet group is listed under the subnet group section
- Public access is no
- Choose existing VPC security groups
- Remove the default security group and add your database security group
- Select your first availability zone as well

## Connectivity



### Virtual private cloud (VPC) [Info](#)

VPC that defines the virtual networking environment for this DB instance.

Demo VPC (vpc-03bd2b389d2a4d45e)

Only VPCs with a corresponding DB subnet group are listed.

**i** After a database is created, you can't change its VPC.

### Subnet group [Info](#)

DB subnet group that defines which subnets and IP ranges the DB instance can use in the VPC you selected.

dbsubnetgroup

### Public access [Info](#)

☐ Yes

Amazon EC2 instances and devices outside the VPC can connect to your database. Choose one or more VPC security groups that specify which EC2 instances and devices inside the VPC can connect to the database.

☒ No

RDS will not assign a public IP address to the database. Only Amazon EC2 instances and devices inside the VPC can connect to your database.

### VPC security group

Choose a VPC security group to allow access to your database. Ensure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing

Choose existing VPC security groups

☐ Create new

Create new VPC security group

### Existing VPC security groups

Choose VPC security groups

MyDatabaseServerSG X

### Availability Zone [Info](#)

us-west-2a

- Scroll down to Additional configuration on the bottom and give it an initial database name and save it in the same spot as your password since it will be used later

- Disable automated backups and encryption since they are not needed (These are normally best practice to leave enabled but the database will spin up faster with those checked off as they are not needed).
- Scroll down all the way to the bottom and create your database

▼ **Additional configuration**

Database options, encryption disabled, backup disabled, backtrack disabled, Enhanced Monitoring disabled, maintenance, CloudWatch Logs, delete protection disabled.

### Database options

Initial database name [Info](#)

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

 ▼

Option group [Info](#)

 ▼

### Backup

☐ **Enable automated backups**  
Creates a point-in-time snapshot of your database

### Encryption

☐ **Enable encryption**  
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

### Monitoring

☐ **Enable Enhanced monitoring**  
Enabling Enhanced monitoring metrics are useful when you want to see how different processes or threads use the CPU.

### Log exports

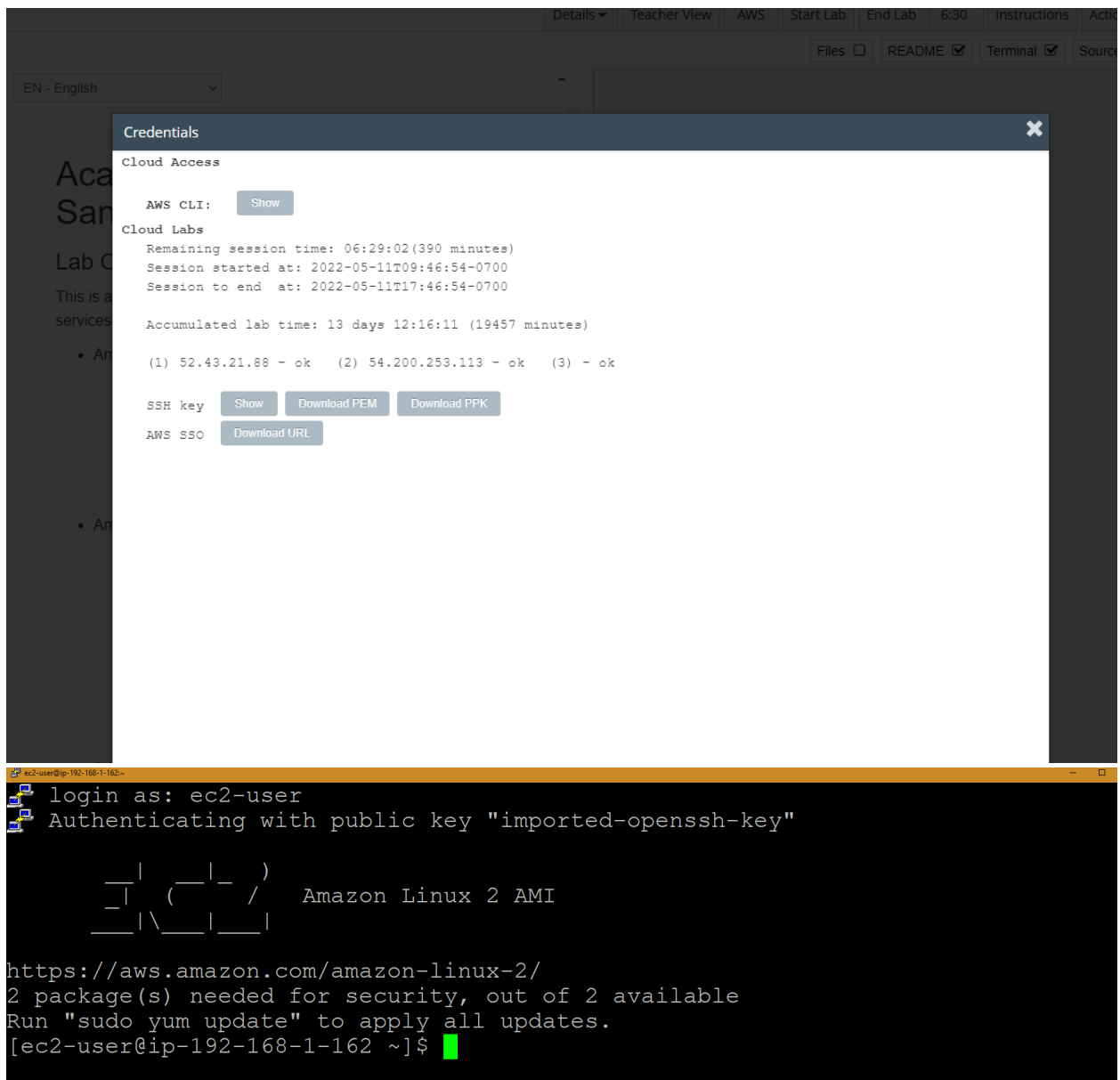
Select the log types to publish to Amazon CloudWatch Logs

☐ **Audit log**

☐ **Error log**

#### Step 4: Test connections

- Ssh into your Bastion Host after downloading both the pem and ppk files from the lab environment



- This is for windows only as I only have a windows machine to work on, sorry mac and linux users
- Go into your powershell and type this command out
  - Pscp -scp -P 22 -i '.\Downloads\labsuser.ppk' -| user ec2-user '.\Downloads\labsuser.pem' ec2-user@bastion-host-public-ip:/home/ec2-user

- ```
PS C:\Users\nickb> Pscpp -scp -P 22 -i '.\Downloads\labsuser.ppk' -l user ec2-user@10.200.253.113:/home/ec2-user
pscp: ec2-user: No such file or directory

labsuser.pem | 1 kB | 1.6 kB/s | ETA: 00:00:00 | 100%
PS C:\Users\nickb>
```

- ```
[ec2-user@ip-192-168-1-162 ~]$ ls
labsuser.pem
[ec2-user@ip-192-168-1-162 ~]$
```

- ```
[ec2-user@ip-192-168-1-162 ~]$ chmod 400 labsuser.pem  
[ec2-user@ip-192-168-1-162 ~]$ ssh -i labsuser.pem ec2-user@192.168.2.172  
The authenticity of host '192.168.2.172 (192.168.2.172)' can't be established.  
ECDSA key fingerprint is SHA256:RURnbNWL6+XNSA3+S9k0FtM1Fy0aAHcv4z7qLtKlm7A.  
ECDSA key fingerprint is MD5:ce:f3:58:79:65:eb:ae:de:6a:3c:51:5c:89:4b:69:88.  
Are you sure you want to continue connecting (yes/no)? yes  
Warning: Permanently added '192.168.2.172' (ECDSA) to the list of known hosts.
```
- \_ | \_ )  
\_ | ( \_ /  
\_ | \ \_ |
- Amazon Linux 2 AMI
- ```
https://aws.amazon.com/amazon-linux-2/  
2 package(s) needed for security, out of 2 available  
Run "sudo yum update" to apply all updates.  
[ec2-user@ip-192-168-2-172 ~]$ ls  
[ec2-user@ip-192-168-2-172 ~]$
```

- Use ping and the private ip address of your web server to ping the web server and see it connect

```
[ec2-user@ip-192-168-2-172 ~]$ ping 192.168.1.252
PING 192.168.1.252 (192.168.1.252) 56(84) bytes of data.
64 bytes from 192.168.1.252: icmp_seq=1 ttl=255 time=0.486 ms
64 bytes from 192.168.1.252: icmp_seq=2 ttl=255 time=0.441 ms
64 bytes from 192.168.1.252: icmp_seq=3 ttl=255 time=0.450 ms
64 bytes from 192.168.1.252: icmp_seq=4 ttl=255 time=0.483 ms
^C
--- 192.168.1.252 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.441/0.465/0.486/0.019 ms
[ec2-user@ip-192-168-2-172 ~]$
```

- Test out connecting to the database by typing out `mysql --user=root -password='Re:Start!9' --host=database-server-endpoint`
- Replace database-server-endpoint with the database server endpoint
- Type show databases; to see your database from the app server

```
[ec2-user@ip-192-168-2-172 ~]$ mysql --user=root --password='Re:Start!9' --host=dbinstance.cxlakalrhkg0.us-west-2.rds.amazonaws.com
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 30
Server version: 10.6.7-MariaDB managed by https://aws.amazon.com/rds/
```

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and other

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
MariaDB [(none)]> show databases;
```

```
+-----+
| Database                |
+-----+
| information_schema      |
| innodb                  |
| mydb                    |
| mysql                   |
| performance_schema     |
| sys                     |
+-----+
6 rows in set (0.00 sec)
```

```
MariaDB [(none)]>
```

- This concludes the lab